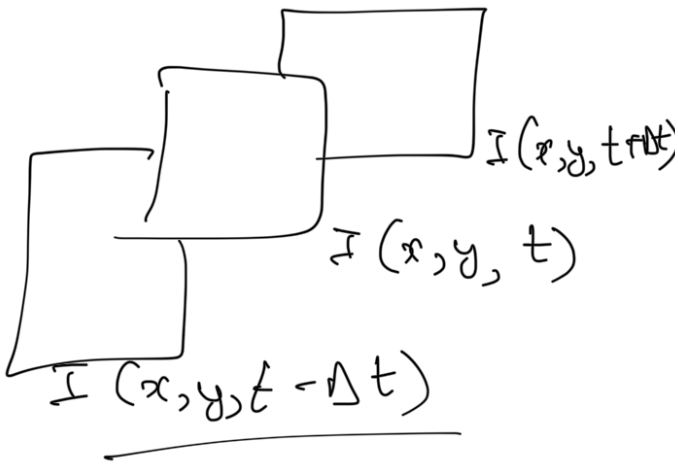
$$I_x u + I_y v + I_t = 0 \Rightarrow \text{OFCE}$$

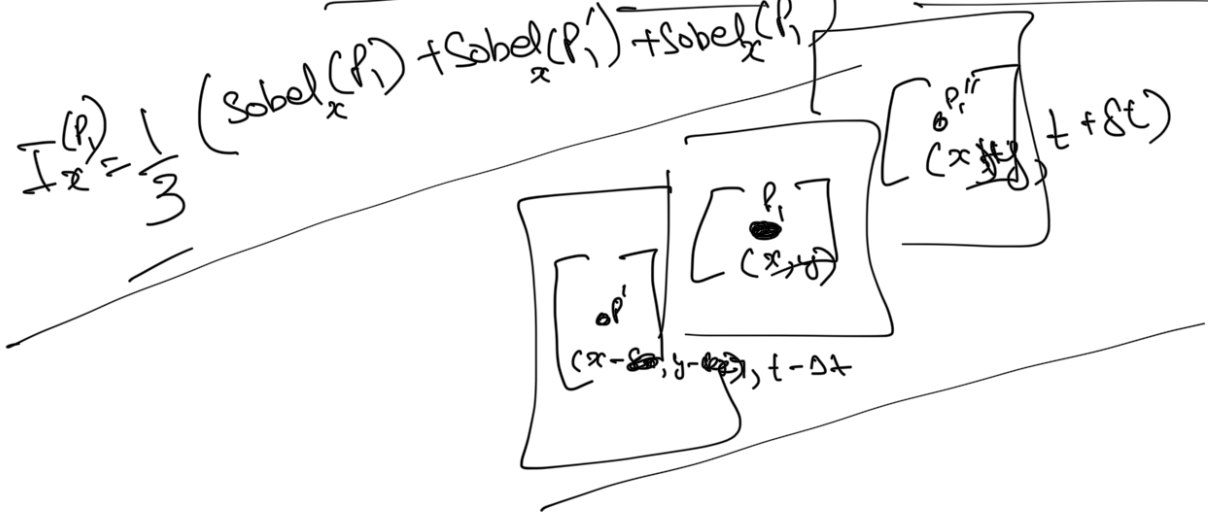
1) Try to increase # of equations
2) Find I_x, I_y and I_t in each equation

Information given $\Rightarrow$



Lucas - Kanade Method

$$I_x^{(P_1)} = \frac{1}{3}\left(\text{Sobel}_x(P_1) + \text{Sobel}_x(P_1') + \text{Sobel}_x(P_1'')\right)$$



$$I_y(P_1) = \frac{1}{3}\left(\text{Sobel}_y(P_1) + \text{Sobel}_y(P_1') + \text{Sobel}_y(P_1''\right)$$

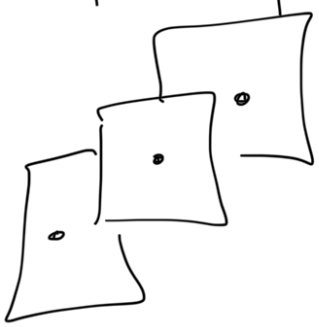
$$\text{Sobel}_x = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}_{3\times 3}$$

Sobely = $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}_{3\times 3}$

$$I_t = \begin{cases} I(x,y,t) - I(x,y,t-\delta t) \\ I(x,y,t+\delta t) - I(x,y,t) \\ \dfrac{I(x,y,t+\delta t) - I(x,y,t-\delta t)}{2} \end{cases}$$

$I_x, I_y, I_t \Rightarrow$ are wrt to a point $P_1(x,y,t)$

$I_{x_{p_1}} u + I_{y_{p_1}} v + I_{t_{p_1}} = 0$ → one equation

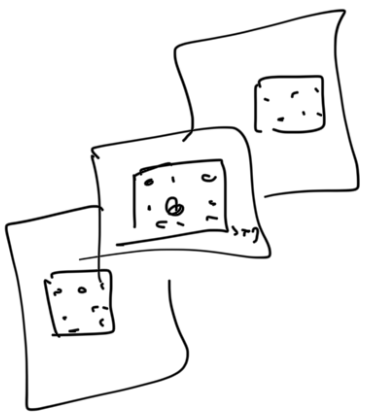


$I_{x_{p_2}} u + I_{y_{p_2}} v + I_{t_{p_2}} = 0$

(u,v) of both P_1 and P_2 are same

→ This can be satisfied if

foreground is detected correctly.



Lucas-Kanade use a neighbourhood region such as (3x3 or 5x5) to increase number of Points

Statistics ⇒ A single sample is never reliable

→ Take as many samples you can.

→ Neighbourhood is usually taken as 3x3 or 5x3

→ Total number of equations is 9 or 25. (5x5) respectively

Under constrained problem

$$I_{x p_1} u + I_{x p_1} v + I_{t p_1} = 0$$
$$I_{x p_2} u + I_{x p_2} v + I_{t p_2} = 0$$

4 equations for 2 variable

$$I_x p_q u + I_x p_q v + I_t p_q = 0$$

↳ Over Constrained problem.

$$\begin{bmatrix} I_x p_1 & I_y p_1 \\ I_x p_2 & I_y p_2 \\ \vdots & \vdots \\ I_x p_q & I_y p_q \end{bmatrix}_{q \times 2} \begin{bmatrix} u \\ v \end{bmatrix}_{2 \times 1} = \begin{bmatrix} -I_t p_1 \\ -I_t p_2 \\ \vdots \\ -I_t p_q \end{bmatrix}_{q \times 1}$$

$$\boxed{A\underline{x} = b}$$



↳ when A is rectangular and equation is overconstrained



↳ Multiply by A^T both sides

$$(A^T A)x = A^T b$$

A^T: $2 \times q$, A: $q \times 2$, $(A^T A)$: 2×2

$$\boxed{\hat{x} = (A^T A)^{-1} A^T b} = \begin{bmatrix} u \\ v \end{bmatrix}$$

Pseudoinverse Solution

=> Where should we be more Careful with this method?

=> Is $(A^TA)_{2\times 2}$ always invertible?

$|A^TA| \neq 0$

A^TA should be full rank

$$\begin{bmatrix} x_1 & x_2 \\ kx_1 & kx_2 \end{bmatrix} \Rightarrow \text{rank} = 1$$

(one row multiple of previous row)

When you are looking at constant regions in image.

When rows of A are not independent

$$\begin{bmatrix} I_{x p_1} & I_{y p_1} \\ I_{x p_2} & I_{y p_2} \\ \vdots & \vdots \\ I_{x p_q} & I_{y p_q} \end{bmatrix}$$

That is why a robust object detection is a necessary pre-processing step in flow estimation

Stick applications:

=> full rank matrix => rank = N

has N eigen values

All eigen are of comparable level.

and

All eigen values are non-zero.

$\lambda_1 \geq \lambda_2$

$\frac{\lambda_1}{\lambda_2}$ should not be very high

$\frac{\lambda_1}{\lambda_2}$ => Condition number of the matrix.

Algorithm

=> Read ~~Hied~~ Video

=> Take 3 snapshots

=> find 9 or 25 equations

=> Solve and find pseudoinverse Solution

=> The solution is the optical flow $[u, v]$

u: velocity in x

v: velocity in y.